

Homework 1

PROBLEM 1 — SOLUTION

Heat into the gas and work done by the gas are positive. The gas is a closed system with boundary work only and negligible changes in bulk kinetic and potential energy. Here E is total internal energy and $e = E/m$. Use $R_u = 8.31446 \text{ J/(mol K)}$; bars denote molar quantities.

(a) (i) **Isothermal:** $T = \text{constant}$.

$$T_{2,\text{iso}} = T_1 = \boxed{300 \text{ K}}, \quad pV = nR_u T_1 = \text{constant}, \quad \frac{V_{2,\text{iso}}}{V_1} = \frac{p_1}{p_2} = \frac{1}{8}.$$

For a calorically perfect gas, $\Delta E = n\bar{c}_v(T_2 - T_1) = 0$. Thus the first law gives $0 = Q - W_{\text{by}}$, or $Q = W_{\text{by}}$. The reversible boundary work is

$$\begin{aligned} W_{\text{by,iso}} &= \int_{V_1}^{V_{2,\text{iso}}} p dV = nR_u T_1 \int_{V_1}^{V_{2,\text{iso}}} \frac{dV}{V} \\ &= nR_u T_1 \ln \frac{V_{2,\text{iso}}}{V_1} = -(1)(8.31446)(300) \ln 8 = \boxed{-5.187 \text{ kJ}}. \end{aligned}$$

(ii) **Reversible adiabatic:** $Q = 0$ and $S_{\text{gen}} = 0$, so $\Delta S = 0$.

$$\begin{aligned} pV^\gamma &= \text{constant}, & \frac{T_{2,\text{ad}}}{T_1} &= \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma}, \\ T_{2,\text{ad}} &= 300(8)^{0.4/1.4} = \boxed{543.4 \text{ K}}. \end{aligned}$$

To obtain the molar heat capacity,

$$\begin{aligned} \bar{c}_p &= \bar{c}_v + R_u, & \gamma &= \frac{\bar{c}_p}{\bar{c}_v}, \\ (\gamma - 1)\bar{c}_v &= R_u, & \bar{c}_v &= \frac{R_u}{\gamma - 1} = 20.7862 \text{ J/(mol K)}. \end{aligned}$$

Apply the first law with $Q = 0$:

$$\Delta E = Q - W_{\text{by,ad}} = -W_{\text{by,ad}},$$

$$W_{\text{by,ad}} = -\Delta E = -m\Delta e = -n\bar{c}_v(T_{2,\text{ad}} - T_1) = \boxed{-5.060 \text{ kJ}}.$$

Unrounded intermediate values are used in the numerical calculations.

(b) Use the ideal-gas entropy relation and convert totals to mass-specific values:

$$\Delta S_{\text{sys}} = n \left(\bar{c}_p \ln \frac{T_2}{T_1} - R_u \ln \frac{p_2}{p_1} \right), \quad \bar{c}_p = \frac{\gamma R_u}{\gamma - 1},$$

$$m = n M_{\text{O}_2} = (1 \text{ mol})(0.03200 \text{ kg/mol}) = 0.03200 \text{ kg}, \quad \Delta e = \frac{\Delta E}{m}, \quad q = \frac{Q}{m}, \quad \Delta s_{\text{sys}} = \frac{\Delta S_{\text{sys}}}{m}.$$

Here $M_{\text{O}_2} = 32.00 \text{ g/mol}$ is the rounded molar mass used for the requested mass-specific results.

(i) **Isothermal:**

$$\Delta E = 0, \quad Q = W_{\text{by,iso}} = -5.187 \text{ kJ},$$

$$\Delta S_{\text{sys}} = n (\bar{c}_p \ln 1 - R_u \ln 8) = -n R_u \ln 8 = -17.29 \text{ J/K},$$

$$\boxed{\Delta e = 0}, \quad \boxed{q = -162.1 \text{ kJ/kg}}, \quad \boxed{\Delta s_{\text{sys}} = -0.5403 \text{ kJ/(kg K)}}.$$

(ii) **Reversible adiabatic:**

$$\Delta E = n \bar{c}_v (T_{2,\text{ad}} - T_1) = +5.060 \text{ kJ}, \quad Q = 0,$$

$$\Delta S_{\text{sys}} = n \left[\frac{\gamma R_u}{\gamma - 1} \frac{\gamma - 1}{\gamma} \ln 8 - R_u \ln 8 \right] = 0,$$

$$\boxed{\Delta e = +158.1 \text{ kJ/kg}}, \quad \boxed{q = 0}, \quad \boxed{\Delta s_{\text{sys}} = 0}.$$

(c) (i) **Isothermal:** The surroundings supply 5.187 kJ of work and receive 5.187 kJ of heat in a 300 K reservoir. For reversible heat transfer, the temperature difference tends to zero.

$$Q_{\text{surr}} = -Q = +5.187 \text{ kJ},$$

$$\Delta S_{\text{surr}} = \frac{Q_{\text{surr}}}{300 \text{ K}} = +17.29 \text{ J/K},$$

$$\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = -17.29 + 17.29 = \boxed{0}.$$

(ii) **Reversible adiabatic:** An ideal work source supplies 5.060 kJ; the insulated boundary permits no heat transfer.

$$\boxed{\Delta S_{\text{sys}} = \Delta S_{\text{surr}} = \Delta S_{\text{universe}} = 0}.$$

The ideal work source transfers energy without entropy. Both processes have zero entropy generation because they are reversible.

(d)

$$W_{\text{in}} = -W_{\text{by}}, \quad \boxed{W_{\text{in,iso}} = 5.187 \text{ kJ}}, \quad \boxed{W_{\text{in,ad}} = 5.060 \text{ kJ}},$$

$$V_{2,\text{iso}} = \frac{n R_u T_1}{p_2} = 3.118 \text{ L}, \quad V_{2,\text{ad}} = \frac{n R_u T_{2,\text{ad}}}{p_2} = 5.648 \text{ L}.$$

For these data, the adiabatic path requires 0.1268 kJ less work input. Its temperature rises, so it reaches 8 bar at a larger volume and stops sooner. The final pressures are equal, but the final states are different; comparing the paths at the same final volume would give a different work comparison.

PROBLEM 5 — SOLUTION

Let h and c denote the initially hot and cold blocks. Each rigid block has constant total heat capacity $C = 500 \text{ J/K}$. The combined system is isolated; kinetic and potential energy changes are negligible.

(a) For the two blocks together, $Q_{\text{ext}} = W_{\text{ext}} = 0$. Hence

$$\begin{aligned}\Delta E_h + \Delta E_c &= 0, \\ C(T_f - 500 \text{ K}) + C(T_f - 300 \text{ K}) &= 0, \\ T_f &= \boxed{400 \text{ K}}.\end{aligned}$$

The hot block loses 50.0 kJ and the cold block gains the same amount.

(b) The actual heat transfer is irreversible. Since entropy is a state property, use a reversible auxiliary path between each block's initial and final states:

$$\delta Q_{\text{rev}} = dE = C dT, \quad \Delta S = \int_{T_i}^{T_f} \frac{\delta Q_{\text{rev}}}{T} = C \ln \frac{T_f}{T_i}.$$

Thus

$$\begin{aligned}\Delta S_h &= (500 \text{ J/K}) \ln \frac{400}{500} = \boxed{-111.57 \text{ J/K}}, \\ \Delta S_c &= (500 \text{ J/K}) \ln \frac{400}{300} = \boxed{+143.84 \text{ J/K}}.\end{aligned}$$

The external surroundings are unchanged, so

$$\boxed{\Delta S_{\text{universe}} = \Delta S_h + \Delta S_c = 32.27 \text{ J/K} > 0.}$$

This is also the entropy generated within the isolated two-block system.

(c) The first law requires only that energy lost by one block equal energy gained by the other; it permits either sign of heat transfer. The second law selects the spontaneous direction. In a uniform-temperature description of each block, let $\delta Q > 0$ flow from the hot block to the cold block. Then

$$dS_{\text{universe}} = -\frac{\delta Q}{T_h} + \frac{\delta Q}{T_c} = \delta Q \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0 \quad (T_h > T_c).$$

Heat flow in the reverse direction, with no other effect, would decrease the universe's entropy and violate the second law.

PROBLEM 7 — SOLUTION

(a) For a single-phase, fixed-composition substance, let h , s , and v be mass-specific properties. Since $h = h(T, p)$,

$$dh = \left(\frac{\partial h}{\partial T} \right)_p dT + \left(\frac{\partial h}{\partial p} \right)_T dp = c_p dT + \left(\frac{\partial h}{\partial p} \right)_T dp.$$

Using $dh = T ds + v dp$ and the mass-specific Gibbs energy $g = h - Ts$,

$$dg = dh - T ds - s dT = (T ds + v dp) - T ds - s dT = -s dT + v dp.$$

$$\left(\frac{\partial g}{\partial T} \right)_p = -s, \quad \left(\frac{\partial g}{\partial p} \right)_T = v.$$

The mixed derivatives of the state function g are equal, so

$$\frac{\partial}{\partial p} \left(\frac{\partial g}{\partial T} \right)_p = \frac{\partial}{\partial T} \left(\frac{\partial g}{\partial p} \right)_T \implies \left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p.$$

Therefore, at fixed T ,

$$\left(\frac{\partial h}{\partial p} \right)_T = T \left(\frac{\partial s}{\partial p} \right)_T + v = v - T \left(\frac{\partial v}{\partial T} \right)_p,$$

and substitution gives

$$\boxed{dh = c_p dT + \left[v - T \left(\frac{\partial v}{\partial T} \right)_p \right] dp.}$$

(b) With the mass-specific gas constant R , $Z = pv/(RT)$ gives

$$v = \frac{ZRT}{p}, \quad \left(\frac{\partial v}{\partial T} \right)_p = \frac{R}{p} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_p \right].$$

Thus,

$$\begin{aligned} v - T \left(\frac{\partial v}{\partial T} \right)_p &= \frac{ZRT}{p} - \frac{RT}{p} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_p \right] \\ &= \frac{ZRT}{p} - \frac{ZRT}{p} - \frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p = -\frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p. \end{aligned}$$

Hence

$$\boxed{dh = c_p dT - \frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p dp.}$$

For an ideal gas, $Z = 1$, so the pressure term vanishes.

(c) The supplied B is molar; use $\bar{v}$, $\bar{h}$, and $R_u = 8.314462618 \text{ J mol}^{-1}\text{K}^{-1}$. Also,

$$\mathcal{M} = 0.0280134 \text{ kg mol}^{-1}, \quad \Delta p = (200 - 1) \times 10^5 = 1.99 \times 10^7 \text{ Pa}.$$

Within the supplied pressure-truncated model,

$$\begin{aligned} Z &= 1 + \frac{B(T)p}{R_u T}, & \bar{v} &= \frac{Z R_u T}{p} = \frac{R_u T}{p} + B(T), \\ \left(\frac{\partial \bar{v}}{\partial T} \right)_p &= \frac{R_u}{p} + B'(T), & B'(T) &\equiv \frac{dB}{dT}. \end{aligned}$$

Since $dT = 0$, the molar form of part (a) gives

$$\begin{aligned} d\bar{h} &= \left[\frac{R_u T}{p} + B - T \left(\frac{R_u}{p} + B' \right) \right] dp = (B - T B') dp, \\ \Delta \bar{h} &= \int_{p_1}^{p_2} (B - T B') dp = (B - T B') \Delta p. \end{aligned}$$

$$\boxed{\Delta h = \frac{\Delta \bar{h}}{\mathcal{M}} = \frac{(B - T B') \Delta p}{\mathcal{M}}}.$$

(i) **Rigid spheres.** With $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$,

$$\begin{aligned} B_{\text{RS}} &= \frac{2\pi}{3} N_A \sigma^3 = \frac{2\pi}{3} (6.02214076 \times 10^{23}) (3.681 \times 10^{-10})^3 \\ &= 6.2908158 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}, \quad B'_{\text{RS}} = 0, \\ \Delta \bar{h}_{\text{RS}} &= (6.2908158 \times 10^{-5}) (1.99 \times 10^7) = 1251.872 \text{ J mol}^{-1}, \\ \Delta h_{\text{RS}} &= \frac{1251.872}{0.0280134} \text{ J kg}^{-1} = \boxed{+44.69 \text{ kJ kg}^{-1}}. \end{aligned}$$

(ii) **Attraction-inclusive model.**

$$\begin{aligned} B - T B' &= [-75.7 - 150(1.01)] \times 10^{-6} = -227.2 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}, \\ \Delta \bar{h} &= (-227.2 \times 10^{-6}) (1.99 \times 10^7) = -4521.280 \text{ J mol}^{-1}, \\ \Delta h &= \frac{-4521.280}{0.0280134} \text{ J kg}^{-1} = \boxed{-161.4 \text{ kJ kg}^{-1}}. \end{aligned}$$

At fixed T , the same model gives

$$\Delta \bar{e} = \Delta \bar{h} - \Delta(p\bar{v}) = (B - T B') \Delta p - B \Delta p = -T B' \Delta p.$$

Rigid spheres have $B' = 0$: internal energy is unchanged, and the positive enthalpy change comes from $p\bar{v}$. Attractions introduce a negative internal-energy contribution; here $B - T B' < 0$, reversing the enthalpy trend.

At 200 bar, however,

$$Z_{\text{RS}} = 2.0088, \quad Z_{\text{attraction}} = -0.21395.$$

The rigid-sphere correction is not small, and the negative Z in model (ii) is unphysical. These are formal results of the assigned truncation; model (ii) fails before the final pressure is reached.

PROBLEM 8 — SOLUTION

Assume steady, isentropic flow of ideal nitrogen, with no heat transfer or shaft work, negligible elevation change, and $V_1 \simeq 0$. Here V denotes flow speed, h is mass-specific enthalpy, and a bar denotes a molar quantity.

$$T_1 = 1200 \text{ K}, \quad \frac{p_2}{p_1} = \frac{1}{50}, \quad M = 0.0280134 \text{ kg/mol}, \quad R_u = 8.314462618 \text{ J/(mol K)}.$$

(a) For the calorically perfect gas (CPG), $\gamma = 1.40$ and $\Delta s = 0$:

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma} \implies T_{2,\text{CPG}} = 1200 \left(\frac{1}{50}\right)^{0.4/1.4} = 392.429 \text{ K}.$$

(b) For the thermally perfect gas (TPG), $\bar{c}_p(T) = 27.0 + 0.0040T$ in J/(mol K). Since $\Delta \bar{s} = 0$,

$$\begin{aligned} 0 &= \int_{T_1}^{T_2} \frac{\bar{c}_p(T)}{T} dT - R_u \ln \frac{p_2}{p_1}, \\ \int_{1200}^{T_2} \left(\frac{27.0}{T} + 0.0040\right) dT &= R_u \ln \frac{1}{50}, \\ 27.0 \ln \frac{T_2}{1200} + 0.0040(T_2 - 1200) &= 8.314462618 \ln \frac{1}{50}. \end{aligned}$$

Solving with MATLAB `fzero` on $300 < T_2 < 500$ K gives

$$T_{2,\text{TPG}} = 404.728 \text{ K}.$$

The left-hand side increases strictly for $T_2 > 0$, since its derivative is $27.0/T_2 + 0.0040 > 0$. Thus the root is unique and satisfies $0 < T_2 < T_1$.

(c) For both models, the positive static enthalpy drop is

$$h_1 - h_2 = \frac{1}{M} \int_{T_2}^{T_1} \bar{c}_p(T) dT, \quad h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \implies V_2 \simeq \sqrt{2(h_1 - h_2)}.$$

CPG:

$$\begin{aligned} \bar{c}_p &= \frac{\gamma R_u}{\gamma - 1} = 29.100619 \text{ J/(mol K)}, \\ h_1 - h_2 &= \frac{\bar{c}_p}{M} (T_1 - T_2) = \frac{29.100619}{0.0280134} (1200 - 392.429) \simeq 838.913 \text{ kJ/kg}, \\ V_{2,\text{CPG}} &= \sqrt{2(838.913 \times 10^3)} \simeq 1295.31 \text{ m/s}. \end{aligned}$$

TPG:

$$\begin{aligned} h_1 - h_2 &= \frac{1}{M} \int_{T_2}^{T_1} (27.0 + 0.0040T) dT \\ &= \frac{27.0(T_1 - T_2) + 0.0020(T_1^2 - T_2^2)}{M} \\ &= \frac{27.0(1200 - 404.728) + 0.0020(1200^2 - 404.728^2)}{0.0280134} \\ &\simeq 857.616 \text{ kJ/kg}, \\ V_{2,\text{TPG}} &= \sqrt{2(857.616 \times 10^3)} \simeq 1309.67 \text{ m/s}. \end{aligned}$$

In each speed calculation, the enthalpy drop is converted to J/kg; $1 \text{ J/kg} = 1 \text{ m}^2/\text{s}^2$.

(d) Both models assume an ideal gas and the same steady, adiabatic, isentropic nozzle flow. CPG uses constant heat capacities; TPG uses temperature-dependent heat capacities. The specified TPG law gives a higher exit temperature but a larger mean heat capacity over its expansion. Its enthalpy drop and exit speed are therefore slightly larger. Neither model includes intermolecular real-gas effects.

AI assistance: ChatGPT assisted with LaTeX preparation and checking for Problem 1, based on my handwritten steps, and with drafting and checking Problems 5, 7, and 8.